

MATH 180 Strategies of Problem Solving

W.R. Casper

Department of Mathematics
California State University Fullerton

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Outline

- 1 SoPS Lecture 2
 - Chomp
 - Strategy Stealing
 - Wrapping up

Outline

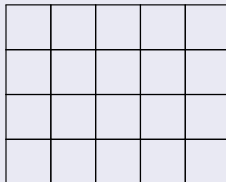
1 SoPS Lecture 2

- Chomp
- Strategy Stealing
- Wrapping up

Chomp

Rules

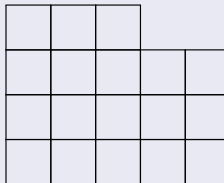
- create a 4×5 square grid (chocolate bar)



- turn: player takes a "bite" of the chocolate bar
- bite: select a square and remove it along with any squares either above or to the right (or both)
- person who eats the (poisoned) last square loses

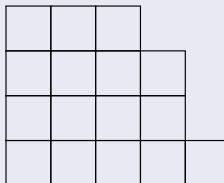
Chomp

Sample turn (P1)



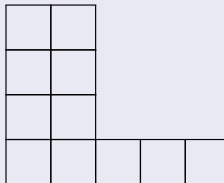
Chomp

Sample turn (P2)



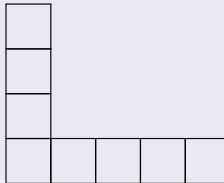
Chomp

Sample turn (P1)



Chomp

Sample turn (P2)



Chomp

Sample turn (P1)



Chomp

Sample turn (P2)



Chomp

Activity (20 mins)

Chomp

Activity (20 mins)

- split into **pairs**

Chomp

Activity (20 mins)

- split into **pairs**
- try playing on a 4×5 board

Chomp

Activity (20 mins)

- split into **pairs**
- try playing on a 4×5 board
- **record data**: did the first or second player win?

Chomp

Activity (20 mins)

- split into **pairs**
- try playing on a 4×5 board
- **record data**: did the first or second player win?
- **record data**: what board states are losing positions?

Chomp

Activity (20 mins)

- split into **pairs**
- try playing on a 4×5 board
- **record data**: did the first or second player win?
- **record data**: what board states are losing positions?
- try to find **patterns**

Chomp: Debrief

Chomp: Debrief

Question

Upload a picture of the interesting losing positions that you found

Chomp: Debrief

Question

Upload a picture of the interesting losing positions that you found

Submit your findings:



Chomp: Debrief

Question

Upload a picture of the interesting losing positions that you found

Submit your findings:



Discussion:

Is it better to be **Player 1** or **Player 2**?

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Vocabulary

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Definition

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A **winning board** is a board state where the next player to play can force a win.

Vocabulary

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A **winning board** is a board state where the next player to play can force a win.

A **losing board** is a board state where the next player to play can be forced to lose.

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Observations

Vocabulary

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Observations

- every board is either a **winning board** or **losing board**

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- every board is either a **winning board** or **losing board**
- **winning board** means there's a move to make it into a **losing board**

Vocabulary

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A **winning board** is a board state where the next player to play can force a win.

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Observations

- every board is either a **winning board** or **losing board**
- **winning board** means there's a move to make it into a **losing board**
- **losing board** means *every* move makes it into a **winning board**

Examples

Losing board



Examples

Winning board



Examples

Losing board



Examples

Winning board



Player 1 Wins!

Player 1 Wins!

Theorem

The starting rectangle is a winning board.

In other words, **Player 1** can guarantee a win with perfect play.

Player 1 Wins!

Theorem

The starting rectangle is a winning board.
In other words, **Player 1** can guarantee a win with perfect play.

Why?

The proof strategy is called a **strategy-stealing argument**.
Assume it's instead a **losing board**.

Strategy Stealing Argument

Strategy Stealing Argument

Proof

Strategy Stealing Argument

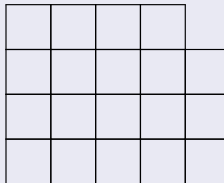
Proof

We get to pick the first move for **Player 1**

Strategy Stealing Argument

Proof

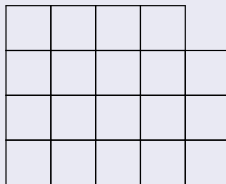
We get to pick the first move for **Player 1**



Strategy Stealing Argument

Proof

We get to pick the first move for **Player 1**



Then this should be a **winning board** for the next to play.

Strategy Stealing Argument

Proof

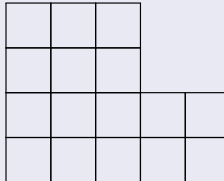
There should be a move turning it into a **losing board**

Strategy Stealing Argument

Proof

There should be a move turning it into a **losing board**

Guess **Player 2**'s move:

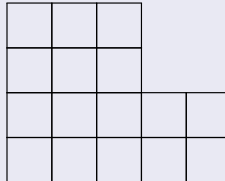


Strategy Stealing Argument

Proof

There should be a move turning it into a **losing board**

Guess **Player 2**'s move:



So if we're right, this is now a **losing board**.

Strategy Stealing Argument

Proof

Restart the game!

Strategy Stealing Argument

Proof

Restart the game!

Have **Player 1** steal **Player 2**'s strategy.

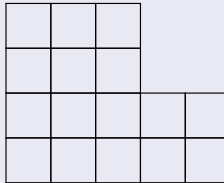
Strategy Stealing Argument

Proof

Restart the game!

Have **Player 1** steal **Player 2's** strategy.

Make **Player 1's** first move be



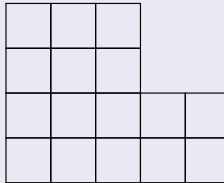
Strategy Stealing Argument

Proof

Restart the game!

Have **Player 1** steal **Player 2**'s strategy.

Make **Player 1**'s first move be



Then now is a losing board for **Player 2**. Impossible!

Strategy Stealing Argument

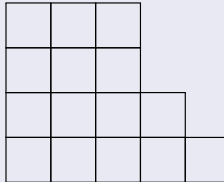
Proof

We must have guessed wrong about **Player 2**'s move.

Strategy Stealing Argument

Proof

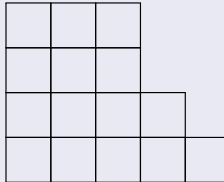
We must have guessed wrong about **Player 2's** move.
Re-guess **Player 2's** move:



Strategy Stealing Argument

Proof

We must have guessed wrong about **Player 2's** move.
Re-guess **Player 2's** move:



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Strategy Stealing Argument

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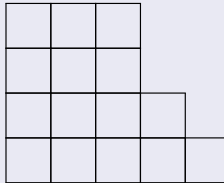
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Have **Player 1** steal **Player 2's** strategy.

Make **Player 1's** first move be



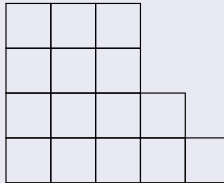
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Make **Player 1**'s first move be



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Strategy Stealing Argument

Proof

Strategy Stealing Argument

Proof

Suppose **Player 2** has a move making a **losing board**.

Strategy Stealing Argument

Proof

Suppose **Player 2** has a move making a **losing board**.
Then **Player 1** could have done that as their first move instead.

Strategy Stealing Argument

Proof

Suppose **Player 2** has a move making a **losing board**.
Then **Player 1** could have done that as their first move instead.
Therefore **Player 1** must be the one able to force a win!

Discussion

Discussion

Question

Does this mean that **Player 1**'s first move should be to take just the upper right square?

Discussion

Question

Does this mean that **Player 1**'s first move should be to take just the upper right square?

Question

Our proof that **Player 1** can force a win was **non-constructive**. We proved a winning strategy exists without finding it. Is this satisfying?

Break time!

Activity (5 minutes)

- get up, stretch get a cup of coffee
- chat up your neighbor, introduce yourself

Perfect Strategy

Perfect Strategy

Question

What is the perfect strategy for 1×7 Chomp?



Perfect Strategy

Question

What is the perfect strategy for 1×7 Chomp?



Question

What is the perfect strategy for $1 \times n$ Chomp?



Perfect Strategy

Perfect Strategy

Activity (20 mins)

Break up into pairs and play Chomp with different board sizes.

Perfect Strategy

Activity (20 mins)

Break up into pairs and play Chomp with different board sizes.
Can you find a perfect strategy for $2 \times n$ Chomp?

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Existence vs. Construction

In a lot of situations, we can prove something exists, long before we figure out how to find it (if ever). If our proof actually shows us how to find it, we call it a **constructive proof**.

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Question 1

How did our proof today compare to our previous proofs in this class?

Discussion

Existence vs. Construction

In a lot of situations, we can prove something exists, long before we figure out how to find it (if ever). If our proof actually shows us how to find it, we call it a **constructive proof**.

Question 1

How did our proof today compare to our previous proofs in this class?

Question 2

Can you think of any situations where knowing something exists could be helpful, even if you don't know how to find it?

Question of the Day

Question of the Day

Puzzle

Start with the number 216.

On each turn, players choose a natural number which divides 216.

Once a number has been chosen, neither it nor any of its divisors may be chosen.

The player who is forced to choose 216 loses.

What's the best first number to choose?

Question of the Day

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On each turn, players choose a natural number which divides 216.

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What's the best first number to choose?

Activity (10 mins)

Discuss the puzzle with your neighbors. What is the solution?

Question of the Day

Question of the Day

Puzzle

This was just 4×4 Chomp!

8	4	2	1
24	12	6	3
72	36	18	9
216	108	54	27

Question of the Day

Puzzle

This was just 4×4 Chomp!

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Activity (10 mins)

Discuss the puzzle with your neighbors. What is the solution?

Final thoughts

Reflection

- Look up the Intermediate Value Theorem in your calculus textbook (or online). What is it proving exists? Why is this proof non-constructive?

Submit your responses by **Tonight!**